

The Fourth HAWC Catalog of Very-High-Energy Gamma-Ray Sources

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We present an updated catalog of TeV gamma-ray sources using data from the 5th reconstruction pass of data from the High Altitude Water Cherenkov Observatory (HAWC). In addition to improved reconstruction and nearly three years of additional data, this new catalog uses a systematic multi-source fitting procedure to model the data with much greater flexibility and accuracy. Besides including more varied modeling of source morphology and spectral shape than previous HAWC catalogs, this catalog uses a robust modeling of galactic diffuse TeV emission through the HERMES package. This catalog reports 85 sources with a Test Statistic greater than 25, including 11 sources not associated with any TevCat source using a distance-based association criterion. Additionally, there are 12 sources not associated with any physical counterpart in Low or High Mass X-ray Binary, the ATNF or Fermi-LAT pulsar, or SNR catalogs of sources. Five of the aforementioned sources have no counterpart in any of the catalogs searched and represent new sources of considerable interest.

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1. Introduction

Accurate γ -ray sky surveys are crucial for studying γ -ray emitters and guiding other instruments towards possible areas of scientific interest. VHE (100 GeV to 100 TeV) and UHE (100+ TeV) γ -ray studies offer insights into areas like dark matter, cosmic-ray acceleration, and other potentially new physics.

While Imaging Atmospheric Cherenkov Telescopes (IACTs) like H.E.S.S. [1], MAGIC [2], VERITAS [3], and CTA [4] provide highly precise measurements for VHE γ -rays, their small Fields of View (FoVs) and low duty cycles make large-scale surveys challenging to undertake.

Extensive Air Shower (EAS) Arrays such as LHAASO [5], Tibet-AS γ [6], ARGO-YBJ [7], and HAWC are more directly suited for TeV γ -ray surveys due to their high duty cycles and large FoVs, allowing them to cover large portions of the sky with high exposure times passively over the course of their operation.

The HAWC observatory, which has been operating for nearly a decade, observes the northern γ -ray sky from hundreds of GeV to hundreds of TeV. This paper introduces an automated Multi-Source Fit (MSF) analysis of HAWC's Pass 5 all-sky data. This new, likelihood ratio-based fitting method for HAWC datasets does not use the method as the previous 2HWC and 3HWC catalogs and performs a blind search, reducing possible biases that older analysis methods may have.

2. The HAWC Detector and Pass 5 Dataset

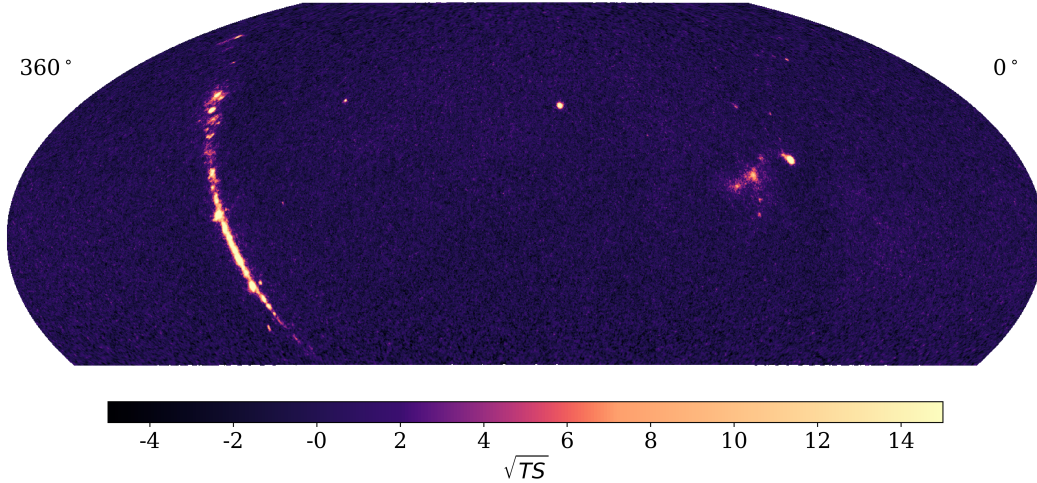


Figure 1: The full all-sky significance map from the HAWC data map used to create the 4HWC. This map is in celestial coordinates, and the significance shown is calculated using a point source assumption with a fixed -2.7 power law index.

The HAWC main array consists of 300 large steel tanks containing 200,000 liters of water each in sealed bladders. The bladders have four PhotoMultiplier Tubes (PMTs) inside, made up

of one large central 10-inch PMT and three smaller 8-inch PMTs arranged in a triangle formation. Charged particles that enter the water in these tanks then produce Cherenkov light. The Cherenkov light is then picked up by the PMTs and recorded. The timing and charge measurements are used to reconstruct the core location and curved shower front of the Extensive Air Shower (EAS) that resulted from an incident highly energetic cosmic-ray or γ -ray. From this reconstructed information, the incoming particle's direction and energy are estimated. The reconstructed data set covers nearly two-thirds of the γ -ray sky (declination -26° to 64°). The details of this process and all the relevant technical details of both the physical array and the newest air shower reconstruction methods can be found in [8] [9]. In addition to the modeling changes, a significant proportion of the improvement of 4HWC over its predecessors in 2HWC and 3HWC stems from the improvement in data quantity and quality of the new data reconstruction. The new data reconstruction in pass 5 has drastically improved the quality of low energy and high zenith events, as well as improved data purity with better gamma/hadron separation across the board. In total the effective area of HAWC at the lowest energies is nearly five times larger, while at the highest zenith, the angular resolution is improved threefold.

3. Source Search Methodology

3.1 The Automated Likelihood Pipeline Search (ALPS)

The Automated Likelihood Pipeline Search (ALPS) method aims to provide a reproducible, automated, and minimally biased pipeline for identifying and fitting γ -ray sources within HAWC data. The pipeline starts by using a local sky map similar to what is shown in Figure 1. The most significant pixel in the smaller Region Of Interest (ROI) is identified and is used as the first seed location. In ROIs that contain a portion of the galactic plane ($b < 10^\circ$), a diffuse emission model is added before any source models are fit. A model is calculated using the default settings of the HERMES computational framework [10]. This framework computes the expected γ -ray emissions from many different radiative processes across the energy range from 100 GeV to 100 TeV. For this work, only the Pion-decay and Inverse Compton (IC) processes were used as the others are expected to be subdominant at these energies. The HERMES framework uses the same cosmic-ray distribution assumed by DRAGON [11] [12] along with the molecular gas distributions from the DAME CO [13] and HI4PI [14] surveys. A template using the predicted γ -ray fluxes across the relevant energy range is subtracted from the data, and a new maximum significance pixel is found. After the diffuse galactic emission is accounted for in regions near the plane, the pipeline proceeds the same way for all ROIs using the maximum significance pixel after any model subtraction. A point source model with a power law spectrum $\frac{dN}{dE} = N_0 \left(\frac{E}{E_0} \right)^{-\alpha}$ where N_0 , E_0 , and α are the flux normalization, pivot energy, and index, respectively, is added at the seed location. The pivot energy is fixed at 2 TeV and represents the value that minimizes the correlation between the flux normalization and index for a source with a Crab-like spectrum at 0° zenith for HAWC. This source is then fit with four free parameters: the Right Ascension (R.A.), Declination (Dec), Flux normalization (N_0), and spectral index (α). The fitting process is done using the maximum likelihood framework described in [15]. 3ML calculates a likelihood value ($\mathcal{L}$) by assuming that the γ -ray events are Poisson distributed. The maximum likelihood parameters are

found by minimizing the negative log ($\mathcal{L}$) of the model parameters. A new significance map of the ROI is made by subtracting the emission from all current models. The process repeats with a new seed location in this new map. The end condition for this process is defined by a Test Statistic $TS = -2 * (\log(\mathcal{L}_{Background}) - \log(\mathcal{L}_{Model}))$. If any new source model fails to reach a TS threshold of 25, then the process terminates. This threshold represents a p-value of approximately 5.03×10^{-5} assuming that Wilks' theorem is verified. After all possible seed sources are added, two phases of hypothesis testing occur. The first substitutes the point-like source morphology for an extended symmetric Gaussian spatial model. During the testing phase, the locations and sizes of all sources except the tested source are fixed. Each source is tested in order of its significance, and a TS threshold of 16 (p-value of 3.17×10^{-5}) is chosen to reject the point-like hypothesis. If any sources have their TS drop below the 25 threshold whenever the new hypothesis is adopted, they are discarded, and nearby sources are refit to find new maximum likelihood parameters. This process is repeated for a second set of hypothesis testing focused on the spectral shape of the sources. Each source has the power law assumption substituted for a log-parabola spectral assumption $\frac{dN}{dE} = N_0 \left(\frac{E}{E_0} \right)^{-\alpha - \beta \log\left(\frac{E}{E_0}\right)}$ which collapses to the power-law assumption when $\beta = 0$. Once all alternative hypotheses are explored, one final refit of all source locations, sizes, and spectral shapes is performed to increase the likelihood of finding the global minimum of the likelihood distribution.

3.2 On- and off-plane ROI selection

The ALPS pipeline is computationally expensive, and each additional source adds to the processing time in a non-linear way. Fitting the full sky or very large ROIs is infeasible given the single-threaded nature of the HAWC 3ML plugin used in this work. To parallelize the fitting process, a series of smaller ROIs that could be analyzed independently was constructed. Two primary approaches were used to divide the sky based on proximity to the galactic plane. For regions near the galactic plane, ROIs 6° in longitude and 20° in latitude are placed along with the plane. These ROIs begin at $l = -1^\circ$ and are placed every 3° until $l = 242^\circ$ with the center at $b = 0^\circ$. This overlapping means each portion of the galactic plane is contained in at least two ROIs, so duplicate sources will need to be accounted for. For any regions more than 10° away from the plane $10^\circ \times 10^\circ$ square ROIs are placed around all pixels with $\sqrt{TS} > 5$. One exception to this is the region near the Geminga and Monogem TeV Halos. These sources are too extended and large to be separated with a single $10^\circ \times 10^\circ$ ROI, so a much larger rectangular ROI was made to contain both sources simultaneously.

3.3 Associating Catalog Sources with Known Sources

To compare the sources found in 4HWC with other astrophysical catalogs an approach modeled after the source combination method in [5] was used. A variable search radius defined by $r_{search} = \sqrt{(0.3^\circ)^2 + (\sigma_{ext})^2 + (\delta_{stat})^2}$ was used. The fixed term (0.3°) is a fixed minimum radius for point like sources with small statistical errors to account for any systematic offsets that may exist between catalogs and instruments. The (σ_{ext}) term allows for large extended sources to look further than point-like sources due to the likelihood of identifying different exact centers for more extended emission. The last term (δ_{stat}) is the statistical uncertainty from the likelihood fit, allowing poorly

localized sources to search further. A maximum of 1° for r_{search} was used to match the 3HWC catalog [16] searching radius.

This variable searching radius is used on two catalogies of astrophysical catalogs. Broadly these sets correspond to TeV catalogs (TeVCat, 1LHAASO [5], 3HWC [16], and HGPS [17]) and catalogs focused on specific source categories (HMXB [18], LMXB [19], ATNF [21], Fermi 3rd Pulsar [20], and SNR [23]). While TeVCat is the most comprehensive TeV source catalog, there are some sources present in the individual catalogs not listed in TeVCat.

3.4 Systematic Uncertainties

The systematic uncertainties for the overall catalog are estimated by taking a subset of ROIs and the sources contained in them. A mix of on-plane, off-plane, low zenith, and high zenith ROIs were used to explore possible effects. These ROIs contain the Crab, Cygnus Cocoon, Galactic Center, Mrk421, Mrk501, and M87.

The position systematics were estimated using the offset of 3 well localized sources by IACT observations: Crab, Mrk501, and M 87. The offset between the 4HWC location and IACT location were added in quadrature to arrive at an uncertainty of 0.04° in R.A. and 0.02° in Declination. Mrk421 was excluded from this uncertainty as another systematic error reduced it's localization accuracy. Despite being an extragalactic source the method found a small extension of 0.09° or approximately 2 pixels in the HAWC Healpix data map. This extension represents a systematic error either on the fitting or data localization side and will be used to estimate the systematic extension uncertainty.

The spectral parameter uncertainty was estimated using a method that uses separate detector response files to simulate systematic changes in the performance of the detector components. The exact changes assumed at given in section 4.5 of [24]. Grouping the files by their category and then adding the most dramatic of each in quadrature and averaging over the sources gives an estimate of the systematics. No declination-dependent or ROI dependent effects appeared to be significant, so these values are used for every source in the catalog. The overall flux measurement systematic uncertainty is $+4.71\%$ and -10.73% . For the spectral index, α , we find an overall systematic of $+4.01\%$ and -4.85% . For the spectral curvature parameter, β , of any log parabola source spectrum models, we find an overall systematic of $+15.38\%$ -8.11% .

3.5 Source Detection Estimation

To best estimate the false positive and negative rates of the ALPS pipeline, sets of simulated HAWC sky maps were computed to test the performance. One set of maps consists of 125 simulated $10^\circ \times 10^\circ$ maps by picking a random R.A. and Declination within HAWCs FoV. Then, one to four sources with randomly chosen sizes and spectra are injected. The background and model emission are then Poisson fluctuated according to the declination of the map. The ALPS pipeline was run on these simulated maps, and the results compared to the injected models. Not all sources were above the expected detection threshold, but of the 125 maps, 97 were able to recover the exact injected model. Of the 239 sources injected, a total of 190 were recovered. Statistically, about 10% of sources were expected to generate with below-sensitivity fluxes for crab-like sources in their declination band and represent true negatives in accordance with detector performance. Other

missed sources are a mix of several situations, like nearby bright sources masking dimmer sources. Or two nearby smaller sources with similar spectral properties are combined into a single larger source. So many true or false negatives occur during this simulated data set, but zero false positives occur. This initially points to the significance thresholds being conservative enough to eliminate false positives in the true data set. However, the number of sub-threshold sources was small in this set of maps, so a second set of 100 maps with no injected sources is used to get a better statistical estimate of the false positive rate. Out of these 100 maps 88 were able to successfully add a model to the Poisson fluctuations and report a TS. This test showed zero sources above 25 TS and 22 above 16 TS. Adjusting for the estimated trials factor of picking the most significant pixel in the $10^\circ \times 10^\circ$ Poisson fluctuated region, the p-value of 16 TS becomes approximately 0.265, which should result in 23.32 sources in 88 maps. Extrapolating the 25 TS threshold to the 128 total ROIs fit for the true catalog estimates a total of 0.5 false positives. However the 128 ROIs are not independent due to significant overlap, so this 0.5 is a conservative upper limit assuming each ROI is totally independent.

3.6 Disambiguating Sources in Overlapping Regions

Section 3.2 brought up the issue that significant double counting will occur in the most source rich region, the galactic plane. To account for this effect, a geometric comparison method was applied to form an association criterion. Each source is convolved with the PSF at its declination to form a circular region associated with the source. Then, each source with some geometric overlap has the overlapping area compared to the total area the sources cover. If this area makes up more than 25% of the total area of the source, then the sources are considered related and are pruned into a single source. The remaining source is the one closest to the center of the ROI, which was fit to reduce any effects of being near the edge of the ROI. A visual explanation is given by Figure 2.

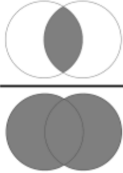
$$\max SF_{\text{overlap}} = \frac{S_{\text{object} \cap \text{source}}}{S_{\text{object} \cup \text{source}}} = \frac{\text{Diagram 1}}{\text{Diagram 2}}$$


Figure 2: A visual representation of the association criterion based upon overlapping circles

4. Results

Compared to the previous 3HWC catalog 4HWC has only 20 additional sources. However the primary method used to produce the 3HWC source list typically adds point sources before larger fixed sources. This can cause regions of extended emission, like the TeV Halo around the Geminga pulsar, to report as several point sources instead of a larger extended source. As such, the increase in the number of sources underestimates the true improvement between the catalogs. In 4HWC, 68 of the total 85 sources report as a symmetric Gaussian at the 4σ level, while in 3HWC, fewer than five reported any extension. The 1LHAASO and 3HWC catalogs only use a simple power law spectral assumption, while 4HWC tests for the spectral curvature at the 4σ level. Eleven new

γ -ray sources are identified that do not associated to any other TeV catalogs using the search radius defined in section 3.3. Associating with the physical catalogs described in section 3.3 shows an overall population distribution. A total of 33 4HWC sources are associated with pulsars, of which 14 are marked as TeV Halo candidates. SNRs represent the next largest category at 22 of the 4HWC sources. X-ray binaries make up the next largest category with 15 sources (7 High mass and 8 Low mass). Three sources are definitively co-located with AGNs, while the remaining 12 sources are unassociated with any of the sources in these five catalogs.

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